

JIYOUN LEE

Email : ljy4694@gmail.com

Github : https://github.com/jiyunni

Website : https://jiyunni.github.io

Phone : +82 10-4934-6279

Education

Dongduk Women's University — Seoul, S.Korea
Mar. 2021 – Feb. 2027 (Expected)

B.S. in Information Statistics

- Double Major in Computer Science
- Total GPA : 3.83/4.5 · Major GPA : 3.88/4.5

Johannes Gutenberg University Mainz — Mainz, Germany
Mar. 2025 – Aug. 2025

Exchange Semester (Computer Science)

- Relevant Coursework : Database Systems, Reinforcement Learning

Research Interests

- Large language models and NLP: user modeling from limited information
- Trustworthy AI: evaluation and reliability of LLM judgments about people
- Recommender systems and personalization: cold-start and user simulation

Publications

- **Lee, J. Y.**, Park, K. H., & Kim, H. H. (2026). "An Adaptive Cross-Domain Recommendation Framework for Structurally Disjoint Domains via Semantic Representation and Similar-User Knowledge Transfer." *Applied Sciences*, 16(19), 9755.
- **Lee, J. Y.**, Park, K. H., & Kim, H. H. (2026). "Cross-Domain Recommendation and Cold-Start Mitigation via Textual Semantic Mapping and Similar-User Transfer." *The Transactions of the Korea Information Processing Society*, 15(6), 508–515. (ASK 2026 Oral)
- Ha, S. J., Oh, S. H., Ban, S. J., **Lee, J. Y.**, & Kim, H. H. (2024). "A Factor Analysis and Regression-Based Prediction Model of Security Deposit Scam Amount for Preventing Rental Scam." *Proceedings of the Annual Symposium of KIPS (ASK 2024)*.

Research & Projects

Adaptive Cross-Domain Recommendation for Structurally Disjoint Domains — Seoul, S.Korea
First Author & Undergraduate Researcher — Mar. 2026 – Aug. 2026

- Linked domains with no shared users or items via SBERT text embeddings to recommend for cold-start users.
- Added an adaptive layer for user-stated preferences, raising Precision@10 by 0.363 (LLM-judged, $p < 0.001$).
- Evaluated with 30 participants (IRB-approved) and LLM judges; agreement was limited (QWK 0.22–0.33).

Personalized Volunteer Activity Recommendation System — Seoul, S.Korea
Team Leader — Sep. 2025 – Dec. 2025

- Recommended activities to cold-start users via SBERT and similar-user transfer, explained by GPT-4o-mini.
- Evaluated with 20 real users: Precision@10 of 0.69 and explanation acceptability of 4.4/5.
- Won the Capstone Design Gold Prize; first-author oral at ASK 2026, later published in the Transactions of KIPS.

Factor Analysis and Prediction of Security Deposit Guarantee Losses — Seoul, S.Korea
Co-author (4th of 5) — Sep. 2023 – Jan. 2024

- Collected, preprocessed, and visualized public housing-price and guarantee-incident data.
- Supported the correlation and regression analysis (cubic model, $R^2 = 0.97$); published at ASK 2024.

Generative AI Platform for Franchise Risk Prediction — Seoul, S.Korea
Team Leader — Sep. 2025 – Nov. 2025

- Built a tool-calling LLM agent (ReAct, LangGraph, MCP) that retrieves evidence and explains franchise risk.
- Predicted next-month risk with LightGBM and extracted store-level risk factors with SHAP.
- Led a 4-person team to the finals of the 13th Big Contest.

Personalized Fashion Recommendation Using Image and Survey Data
Seoul, S.Korea
Aug. 2024 – Nov. 2024

Team Leader

- Cropped person regions with YOLO and extracted style features with a ResNet18 classifier.
- Selected predictive survey items via chi-square tests and Cramér's V, combined in item-based CF.
- Achieved 92.11% accuracy (F1 90.37%), best with survey weighted most (0.6); Excellence Award, 2024 DCC.

Individualized Housing Recommendation Service (3rd Open Data Forum)
Seoul, S.Korea
Oct. 2024 – Nov. 2024

Team Member

- Co-developed a housing recommender for single-person households using public-data living indices.
- Designed the type-specific weighted ranking over survey-based preference types; selected as a finalist.

Strawberry Smart Farm Control Using Reinforcement Learning
Mainz, Germany
May 2025 – Aug. 2025

Individual Project

- Built a custom Gymnasium environment for temperature, humidity, and lighting, and compared PPO and SAC.

Internships

Quad Miners (R&D Intern)
Seoul, S.Korea
Aug. 2026 – Feb. 2027 (Expected)

- Supporting a government-funded AI security research project for defense applications.
- Surveying AI security technology trends and participating in AI model development.

WPP Media Korea (Data Analysis Intern)
Seoul, S.Korea
Jan. 2026 – Feb. 2026

- Analyzed advertising performance data for luxury-brand campaigns and identified key performance trends.
- Provided data-driven recommendations for future campaign strategy and execution.

Awards & Honors

Awards

2026.01	**Gold Prize,** 2025-2 Capstone Design Collaborative Learning Competition	Dongduk Women's Univ
2025.01	**Excellence Project Award,** Outlier (Academic Club) Data Visualization Project Showcase	Dongduk Women's Univ
2024.12	**Excellence Award**, 2024 Data Creator Camp (DCC) Undergraduate Category	K-ICT Big Data Center
2024.06	**Encouragement Award**, The 2nd Generative AI Ideathon	Wrtn Technologies Inc
2024.01	**Excellence Project Award**, Outlier (Academic Club) Data Analysis Project Showcase	Dongduk Women's Univ
2023.08	**Excellence Award**, The 4th National University Student Sharing & Volunteer Hackathon	Korea Red Cross

Honors

2024	Semester High Honors, Awarded for outstanding academic performance (Fall Semester)	Dongduk Women's Univ
2024	Semester High Honors, Awarded for outstanding academic performance (Spring Semester)	Dongduk Women's Univ
2023	Semester High Honors, Awarded for outstanding academic performance (Spring Semester)	Dongduk Women's Univ

Extracurricular Activities

Outlier (Academic Data Society)
Seoul, S.Korea
Sep. 2023 – Jan. 2025

Data Analyst 6th & 7th

- Participated in weekly data analysis projects on predictive modeling, visualization, and presentation.

BDA (Big Data Analysis Academic Society)
Seoul, S.Korea
Mar. 2024 – Sep. 2024

Data Pre-processing Track

- Completed an intensive curriculum with weekly study sessions and hands-on assignments.
- Specialized in data cleaning and feature engineering using Python to handle real-world datasets.

Teaching

CJ UNIT (11th Gen), CJ OliveNetworks
Seoul, S.Korea
Aug. 2024 – Dec. 2024

SW Education Volunteer

- Taught weekly software coding classes for a semester to students facing regional educational gaps.

Mathematics Mentor
Ansan, S.Korea
Jul. 2023 – Sep. 2023

Ansan Youth Shelter

- Provided personalized mathematics tutoring and academic mentoring for underprivileged youth at a residential shelter.

Certifications & Skills

Certificate

- 2024.09 - SQLD

Skills

- **Programming** Python, R, SQL, C, Java
- **Machine Learning** PyTorch, TensorFlow, scikit-learn, Sentence-Transformers, LLM APIs
- **Data Analysis** Pandas, NumPy, Matplotlib, Tableau